PROJ OF A GRADED RING IS A SCHEME WITH QUASI-COHERENT TILDES

THE STACKS PROJECT AUTHORS

Editorial note. 'In exactly the same way we show that M-tilde is quasi-coherent' mirrors the O_X argument with M_(f) in place of S_(f); the key isomorphism for M-tilde over D_+(f) is displayed.

Allowed standard prerequisites (editorial).

- Standard opens and their algebra. lemma-standard-open (carded this batch). Ring identifications S_(fg) = (S_(f))_(g) and the restriction maps between degree-zero parts.
- Chart openings inside Spec(S_(f)). lemma-standard-open-proj (carded this batch). Every standard open of Spec(S_(f)) comes from a D_+(h) inside D_+(f).
- Point homeomorphism. algebra-lemma-topology-proj (cross-chapter, named). The homeomorphism from D_+(f) to Spec(S_(f)).
- Sections over standard opens. lemma-proj-sheaves (same chapter, section-level display, not carded). Sections of O and of M-tilde over D_+(f) are S_(f) and M_(f).

Source reference. Original lemma 01M7: sources/stacks-constructions/pinned-original-source.tex, source lines 1400–1400. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01M4: sources/stacks-constructions/pinned-original-source.tex, source lines 1172–1172. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01MA: sources/stacks-constructions/pinned-original-source.tex, source lines 1466–1466. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 01HR: focused-seeds/geometry/S9-cons-proj-scheme/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:constructions.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *Let S be a graded ring. The locally ringed space $Proj(S)$ is a scheme. The standard opens $D_+(f)$ are affine opens. For any graded S-module M the sheaf $\widetilde{M}$ is a quasi-coherent sheaf of $\mathcal{O}_{Proj(S)}$-modules.*

Proof. Consider a standard open $D_+(f) \subset \text{Proj}(S)$. By Lemmas 01M4 and 01M7 we have $\Gamma(D_+(f), \mathcal{O}_{\text{Proj}(S)}) = S_{(f)}$, and we have a homeomorphism $\varphi : D_+(f) \to \text{Spec}(S_{(f)})$. For any standard open $D(g) \subset \text{Spec}(S_{(f)})$ we may pick an $h \in S_+$ as in Lemma 01MA. Then $\varphi^{-1}(D(g)) = D_+(h)$, and by Lemmas 01M7 and 01M4 we see

$$\Gamma(D_+(h), \mathcal{O}_{\text{Proj}(S)}) = S_{(h)} = (S_{(f)})_{h^{\deg(f)}/f^{\deg(h)}} = (S_{(f)})_g = \Gamma(D(g), \mathcal{O}_{\text{Spec}(S_{(f)})}).$$

Thus the restriction of $\mathcal{O}_{\text{Proj}(S)}$ to $D_+(f)$ corresponds via the homeomorphism φ exactly to the sheaf $\mathcal{O}_{\text{Spec}(S_{(f)})}$ as defined in Schemes, Section 01HR. We conclude that $D_+(f)$ is an affine scheme isomorphic to $\text{Spec}(S_{(f)})$ via φ and hence that $\text{Proj}(S)$ is a scheme.

In exactly the same way we show that $\widetilde{M}$ is a quasi-coherent sheaf of $\mathcal{O}_{\text{Proj}(S)}$-modules. Namely, the argument above will show that

$$\widetilde{M}|_{D_+(f)} \cong \varphi^* \left(\widetilde{M_{(f)}} \right)$$

which shows that $\widetilde{M}$ is quasi-coherent. □